

Training village PCBs documentation

2026-03-03

1 Introduction

The system is comprised of three boards. The main one is a raspberry pi shield, seen in fig 1, which exposes three ethernet connectors that go to the other two boards. It also features a DC connector to power the servos and other peripherals.

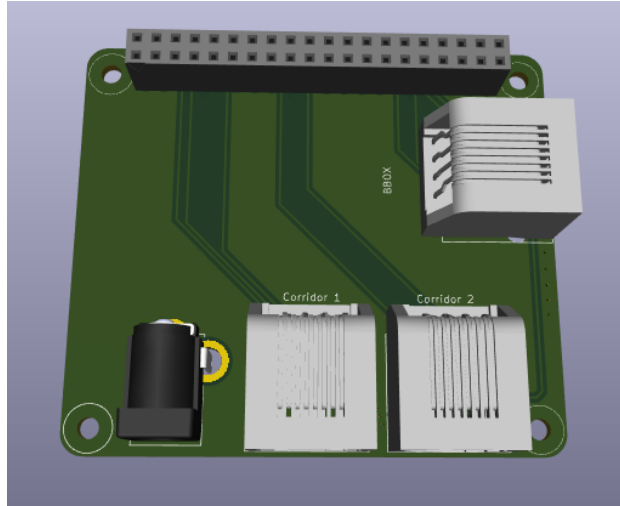


Figure 1: 3D render of the Raspberry Pi shield.

To control the corridor a PCB is designed, this board has connectors to control the motors, the LED strip, the

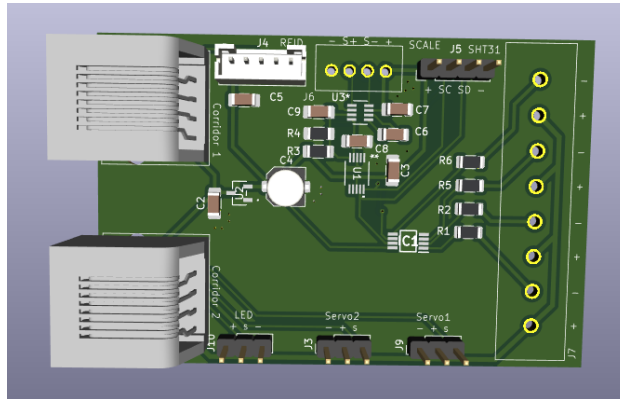


Figure 2: 3D render of the corridor PCB.

The behaviour box is controlled from another board that features one connector for a servo and two connectors for LED strips. This board also allows to connect IR LEDs with a screwed connector.

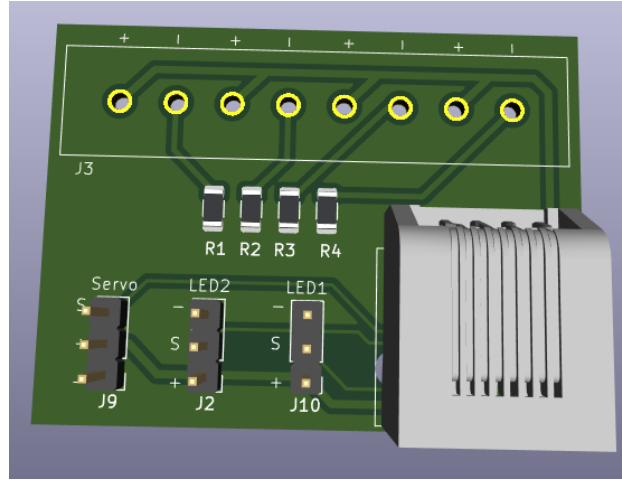


Figure 3: 3D render of the behaviour box PCB.

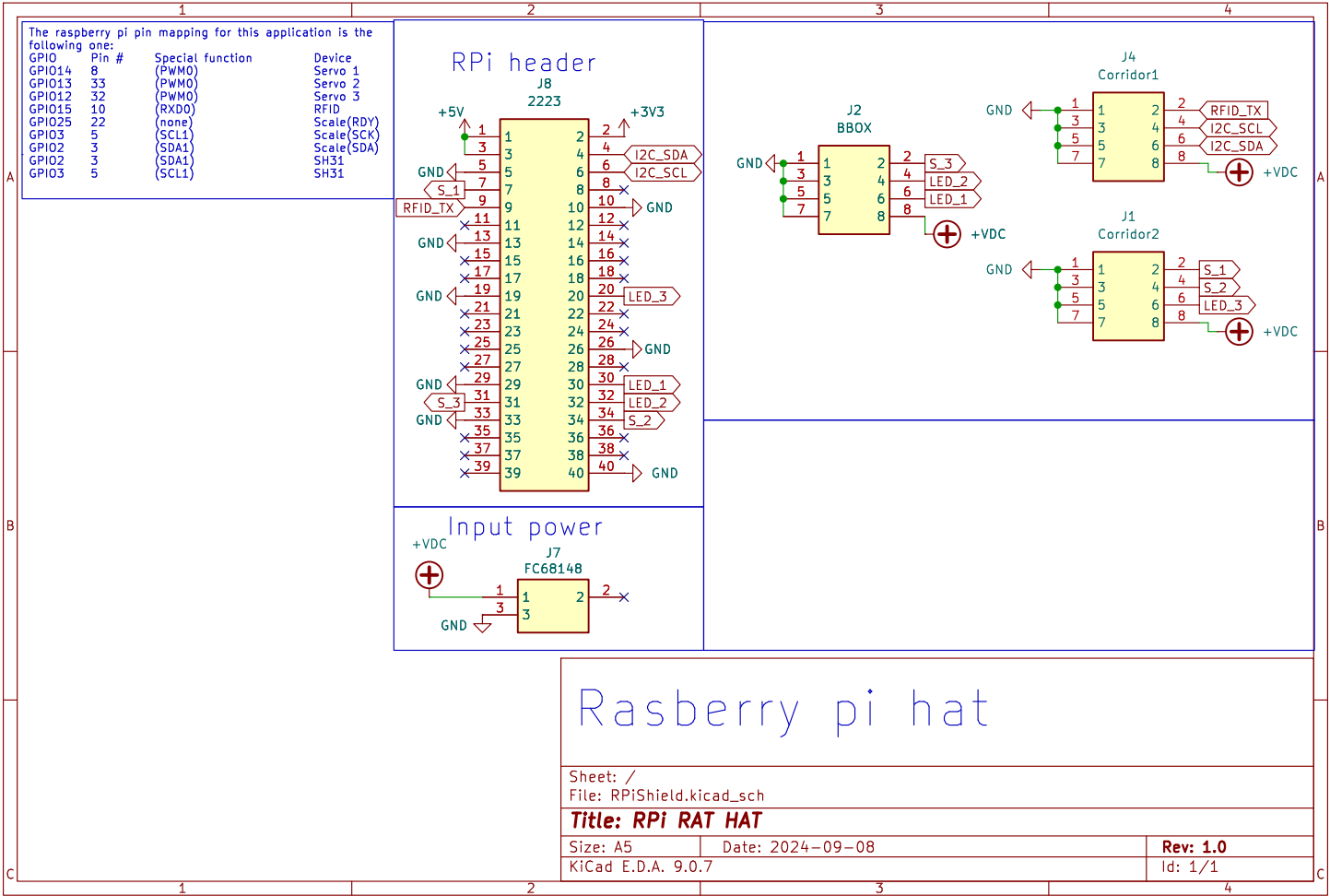
2 Absolute maximum ratings

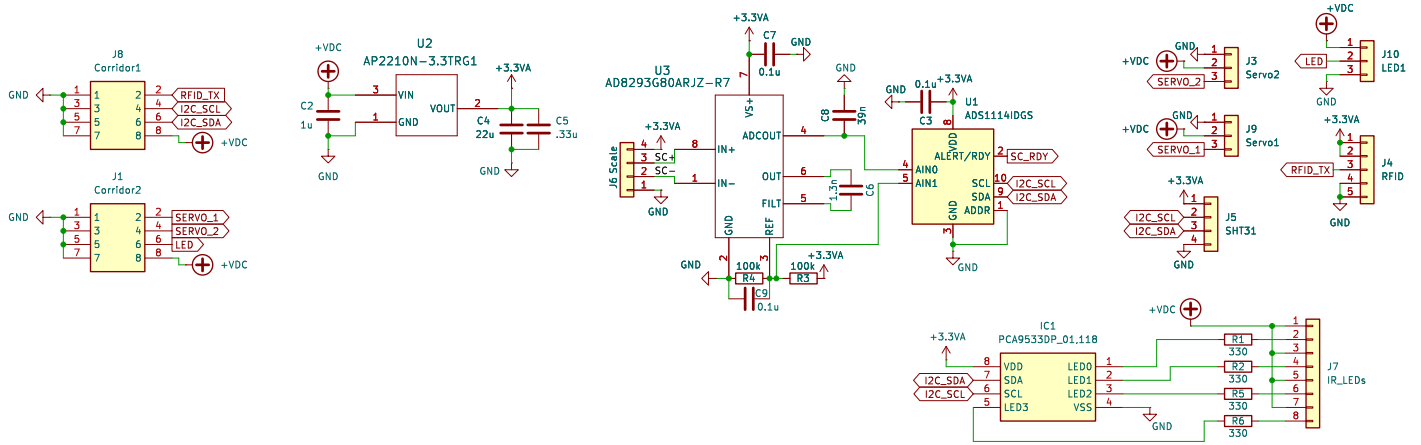
Here is presented a table that summarizes the absolute maximum rating of the system.

<i>Parameter</i>	<i>Value</i>	<i>Units</i>
Input DC jack voltage	0-6	V
Output current	1	A

Table 1: Absolute maximum ratings table

3 Annex





Corridor

Sheet: /
File: Corridor.kicad_sch

Title:

Size: A4
KiCad E.D.A. 9.0.7

Date:

Rev:

Id: 1/1

